

APPLICATION TO FINANCIAL FRICTIONS

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Threshold Models with Three Regimes

Another extension is consider threshold models with three or more regimes

$$y_i = \begin{cases} \beta_1 x_i + e_i & \text{if } q_i \leq \gamma_1 \\ \beta_2 x_i + e_i & \text{if } \gamma_1 < q_i \leq \gamma_2 \\ \beta_3 x_i + e_i & \text{if } q_i > \gamma_2 \end{cases}$$

or in a compact form,

$$y_i = \beta_1 x_i 1(q_i \leq \gamma_1) + \beta_2 x_i 1(\gamma_1 < q_i \leq \gamma_2) + \beta_3 x_i 1(q_i > \gamma_2) + e_i, \quad (1)$$

Where

- q_i is the threshold variable assumed to be exogenous and must have a continuous distribution;
- x_i is a set of regressors assumed to be exogenous;
- γ are the threshold parameters that defines three regimes;
- $\beta = (\beta_1, \beta_2, \beta_3)'$ are the slope parameters.

Threshold Models with Three Regimes

Estimation

Using a vector notation, the model can be written as

$$Y = X(\gamma)' \beta + e, \quad (2)$$

where,

$$X(\gamma) = \begin{bmatrix} x_1 1(q_1 \leq \gamma_1) & x_1 1(\gamma_1 < q_1 \leq \gamma_2) & x_1 1(q_1 > \gamma_2) \\ x_2 1(q_2 \leq \gamma_1) & x_2 1(\gamma_1 < q_2 \leq \gamma_2) & x_2 1(q_2 > \gamma_2) \\ x_3 1(q_3 \leq \gamma_1) & x_3 1(\gamma_1 < q_3 \leq \gamma_2) & x_3 1(q_3 > \gamma_2) \\ \vdots & \vdots & \vdots \\ x_n 1(q_n \leq \gamma_1) & x_n 1(\gamma_1 < q_n \leq \gamma_2) & x_n 1(q_n > \gamma_2) \end{bmatrix},$$

and $\beta = (\beta_1, \beta_2, \beta_3)'$ and $\gamma = (\gamma_1, \gamma_2)'$

Threshold Models with Three Regimes

Estimation

Standard estimation procedure follows the next steps

- 1 Form a grid on the empirical support of the threshold variable q_i .
- 2 For each value of $\gamma = (\gamma_1, \gamma_2)'$ on this grid, the conditional least square (CLS) estimator for $\beta = (\beta_1, \beta_2, \beta_3)'$ is

$$\hat{\beta}(\gamma) = (X^*(\gamma)'X^*(\gamma))^{-1}X^*(\gamma)'Y^*.$$

- 3 The vector of regression residuals is

$$\hat{e}^*(\gamma) = Y^* - X^*(\gamma)\hat{\beta}(\gamma).$$

- 4 The estimator of γ is the value that minimizes the sum of squared errors

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \hat{e}^*(\gamma)' \hat{e}^*(\gamma) \equiv \underset{\gamma}{\operatorname{argmin}} S(\gamma).$$

- 5 Once $\hat{\gamma}$ is obtained, the slope coefficient estimate is $\hat{\beta} = \hat{\beta}(\hat{\gamma})$.

Threshold Models with Three Regimes

Estimation

But using a grid search method is cumbersome to implement in practice. Then, Hansen (1999) proposes to estimate the model sequentially, that is:

- 1 Estimate as if there is one threshold parameter, say γ_1 . Then $\hat{\gamma}_1$ is obtained by minimizing

$$\hat{\gamma}_1 = \underset{\gamma_1}{\operatorname{argmin}} S(\gamma_1).$$

- 2 Fixing the first-stage estimate $\hat{\gamma}_1$, the second-stage estimator of γ_2 is

$$\hat{\gamma}_2^r = \underset{\gamma_2}{\operatorname{argmin}} S(\hat{\gamma}_1, \gamma_2).$$

- 3 Fixing the second-stage estimate $\hat{\gamma}_2^r$, the refinement first threshold estimator is

$$\hat{\gamma}_1^r = \underset{\gamma_1}{\operatorname{argmin}} S(\gamma_1, \hat{\gamma}_2^r).$$

- 4 Once $\hat{\gamma}^r = (\hat{\gamma}_1^r, \hat{\gamma}_2^r)$ is obtained, the slope estimator is $\hat{\beta} = \hat{\beta}(\hat{\gamma}^r)$.

Threshold Models with Three Regimes

Testing

In the context of a threshold model with three regimes, there are either no thresholds, one threshold, or two threshold parameters.

Previously, [Hansen \(2000\)](#) introduced a test of no thresholds against one threshold, and suggested a bootstrap to approximate the asymptotic p -value.

In the context of model (1) we need a further test to discriminate between one and two thresholds.

[Hansen \(1999\)](#) approximates a likelihood ratio test of the null hypothesis of one threshold versus the alternative hypothesis of two thresholds in the model, where a bootstrap procedure to approximate the sampling distribution is suggested.

Application 2: Investment and Financing Constraints

- Classical models of the firm assume the existence of perfect financial markets on which firms can borrow the needed resources for investment projects.
- Alternative models of financing place restrictions on the extent of external financing.
- The presence of financing constraints implies that a firm's cash flow will be positively related to its investment rate only when the firm faces constraints on external financing ([Fazzari et al., 1988](#)).
- A natural candidate of threshold variable is the existing debt level, because it seems reasonable to believe that banks will be reluctant to lend money to debt-heavy firms.

Application 2: Investment and Financing Constraints

Hansen (1999) estimates the following multiple threshold regression model

$$\begin{aligned} I_{it} = & \mu_i + \theta_1 Q_{it-1} + \theta_2 Q_{it-1}^2 + \theta_3 Q_{it-1}^3 \\ & + \theta_5 Q_{it-1} D_{it-1} + \theta_4 D_{it-1} + \beta'_1 CF_{it-1} 1(q_{it} \leq \gamma_1) \\ & + \beta'_2 CF_{it-1} 1(\gamma_1 < q_{it} \leq \gamma_2) + \beta'_3 CF_{it-1} 1(q_{it} > \gamma_2) + e_{it}, \quad (3) \end{aligned}$$

where

- I_{it} is the investment to assets ratio,
- Q_{it} is the total market value to assets ratio,
- D_{it} is the long-term debt to assets ratio,
- CF_{it} is the cash flow to assets ratio,
- i indexes firms and t indexes time.

Application 2: Investment and Financing Constraints

Table 1: Tests for threshold effects

Test for single threshold	
F_1	32.6
P-value	0.003
(10%, 5%, 1% critical values)	(12.4, 14.8, 26.2)
Test for double threshold	
F_2	25.8
P-value	0.017
(10%, 5%, 1% critical values)	(12.3, 14.9, 42.9)
Test for triple threshold	
F_3	4.2
P-value	0.723
(10%, 5%, 1% critical values)	(10.9, 13.3, 22.9)

Application 2: Investment and Financing Constraints

Table 2: Asymptotic confidence interval in threshold model

	Threshold estimate (%)	95% confidence interval	99% confidence interval
$\hat{\gamma}_1$	0.0157	[0.0139, 0.0181]	[0.0120, 0.0239]
$\hat{\gamma}_2$	0.5362	[0.5305, 0.5629]	[0.5190, 0.5693]

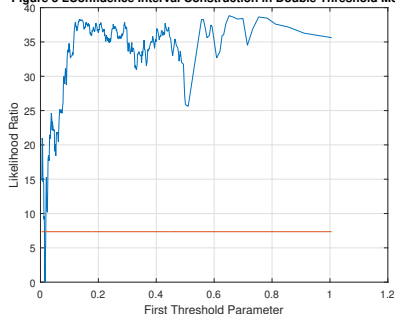
Note: Asymptotic critical values are reported in Hansen (2000).

Application 2: Investment and Financing Constraints

Figure 1: Confidence interval construction for thresholds

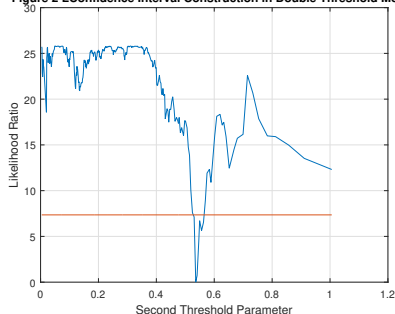
(a) First threshold parameter

Figure 3 LConfidence Interval Construction in Double Threshold Model



(b) Second threshold parameter

Figure 2 LConfidence Interval Construction in Double Threshold Model



Application 2: Investment and Financing Constraints

Table 3: Estimation results - Dependent variable: Investment

Regressor	Coefficient estimate	OLS SE	White SE
Q_{it}	0.010	0.001	0.002
$Q_{it}^2/10^3$	-0.198	0.026	0.064
$Q_{it}^3/10^6$	1.047	0.199	0.448
D_{it}	-0.016	0.005	0.009
$Q_{it}D_{it}$	-0.001	0.001	0.002
$CF_{it}1(D_{it} \leq 0.0125)$	0.063	0.006	0.014
$CF_{it}1(0.0125 < D_{it} \leq 0.5362)$	0.098	0.006	0.010
$CF_{it}1(0.5362 > D_{it})$	0.039	0.012	0.031

Threshold Models with Two Threshold Variables

Lamadrid Contreras & Ramírez-Rondán (2023) consider a model with two threshold variables

$$y_{it} = \begin{cases} \mu_i + \beta_1 x_{it} + e_{it} & \text{if } q_{1it} \leq \gamma_1, q_{2it} \leq \gamma_2 \\ \mu_i + \beta_2 x_{it} + e_{it} & \text{if } q_{1it} \leq \gamma_1, q_{2it} > \gamma_2 \\ \mu_i + \beta_3 x_{it} + e_{it} & \text{if } q_{1it} > \gamma_1, q_{2it} \leq \gamma_2 \\ \mu_i + \beta_4 x_{it} + e_{it} & \text{if } q_{1it} > \gamma_1, q_{2it} > \gamma_2, \end{cases}$$

or in a compact form,

$$y_{it} = \mu_i + \beta_1' x_{it} 1(q_{1it} \leq \gamma_1, q_{2it} \leq \gamma_2) + \beta_2' x_{it} 1(q_{1it} \leq \gamma_1, q_{2it} > \gamma_2) + \beta_3' x_{it} 1(q_{1it} > \gamma_1, q_{2it} \leq \gamma_2) + \beta_4' x_{it} 1(q_{1it} > \gamma_1, q_{2it} > \gamma_2) + e_{it} \quad (4)$$

Where

- $q_{it} = q_{1it}, q_{2it}$ are the threshold variable assumed to be exogenous and must have a continuous distribution;
- $\gamma = (\gamma_1, \gamma_2)'$ are the threshold parameters;
- $\beta = (\beta_1, \beta_2, \beta_3, \beta_4)'$ are the slope parameters.

Threshold Models with Two Threshold Variables

Another compact representation is to set

$$x_{it}(\gamma) = \begin{pmatrix} x_{it}1(q_{1it} \leq \gamma_1, q_{2it} \leq \gamma_2) \\ x_{it}1(q_{1it} \leq \gamma_1, q_{2it} > \gamma_2) \\ x_{it}1(q_{1it} > \gamma_1, q_{2it} \leq \gamma_2) \\ x_{it}1(q_{1it} > \gamma_1, q_{2it} > \gamma_2) \end{pmatrix}$$

Then, equation (4) equals

$$y_{it} = \mu_i + \beta' x_{it}(\gamma) + e_{it}. \quad (5)$$

One traditional method to eliminate the individual fixed specific effect μ_i is to remove individual-specific means

$$\bar{y}_i = \mu_i + \beta' \bar{x}_i(\gamma) + \bar{e}_i, \quad (6)$$

Threshold Models with Two Threshold Variables

where $\bar{y}_i = T^{-1} \sum_{t=1}^T y_{it}$, $\bar{e}_i = T^{-1} \sum_{t=1}^T e_{it}$ and

$$\begin{aligned}\bar{x}_i(\gamma) &= \frac{1}{T} \sum_{t=1}^T x_{it}(\gamma) \\ &= \begin{pmatrix} \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{1it} \leq \gamma_1, q_{2it} \leq \gamma_2) \\ \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{1it} \leq \gamma_1, q_{2it} > \gamma_2) \\ \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{1it} > \gamma_1, q_{2it} \leq \gamma_2) \\ \frac{1}{T} \sum_{t=1}^T x_{it} 1(q_{1it} > \gamma_1, q_{2it} > \gamma_2) \end{pmatrix}\end{aligned}$$

Taking the difference between (5) and (6) yields

$$y_{it}^* = \beta' x_{it}^*(\gamma) + e_{it}^*, \quad (7)$$

where $y_{it}^* = y_{it} - \bar{y}_i$, $x_{it}^*(\gamma) = x_{it}(\gamma) - \bar{x}_i(\gamma)$ and $e_{it}^* = e_{it} - \bar{e}_i$.

Threshold Models with Two Threshold Variables

Estimation

Using vector notation, standard estimation procedure follows the next steps

- 1 Form a grid on the empirical support of the threshold variable q_{it} .
- 2 For each value of γ on this grid, the conditional least square (CLS) estimator for β is

$$\hat{\beta}(\gamma) = (X^*(\gamma)'X^*(\gamma))^{-1}X^*(\gamma)'Y^*.$$

- 3 The vector of regression residuals is

$$\hat{e}^*(\gamma) = Y^* - X^*(\gamma)\hat{\beta}(\gamma).$$

- 4 The estimator of γ is the value that minimizes the sum of squared errors

$$\hat{\gamma} = \underset{\gamma}{\operatorname{argmin}} \hat{e}^*(\gamma)' \hat{e}^*(\gamma) \equiv \underset{\gamma}{\operatorname{argmin}} S(\gamma).$$

- 5 Once $\hat{\gamma}$ is obtained, the slope coefficient estimate is $\hat{\beta} = \hat{\beta}(\hat{\gamma})$.

Threshold Models with Two Threshold Variables

Estimation

We propose to estimate the model sequentially, that is:

- 1 Estimate as if there is one threshold variable, say q_{1it} . That is, assume that $\beta'_1 = \beta'_2$ and $\beta'_3 = \beta'_4$. Then $\hat{\gamma}_1$ is obtained by minimizing

$$\hat{\gamma}_1 = \underset{\gamma_1}{\operatorname{argmin}} S(\gamma_1).$$

- 2 Fixing the first-stage estimate $\hat{\gamma}_1$, the second-stage estimator of γ_2 is

$$\hat{\gamma}_2^r = \underset{\gamma_2}{\operatorname{argmin}} S(\hat{\gamma}_1, \gamma_2).$$

- 3 Fixing the second-stage estimate $\hat{\gamma}_2^r$, the refinement first threshold estimator is

$$\hat{\gamma}_1^r = \underset{\gamma_1}{\operatorname{argmin}} S(\gamma_1, \hat{\gamma}_2^r).$$

- 4 Once $\hat{\gamma}^r = (\hat{\gamma}_1^r, \hat{\gamma}_2^r)$ is obtained, the slope estimator is $\hat{\beta} = \hat{\beta}(\hat{\gamma}^r)$.

Application 3: Investment and Financing Constraints

- If there are financial constraints, firms investment decisions are not independent to fluctuations of internal finance (cash flow).
- Empirical studies divide firms into groups of constrained and unconstrained firms based on a variable that measures capital market imperfections or financial constraints ([Schiantarelli, 1996](#); [Hubbard, 1998](#)).
- [Fazzari et al. \(1988\)](#) argue that cash flow and investment are positively related only when the firm faces constraints on external financing.
- [Hansen \(1999\)](#) and [Hu and Schiantarelli \(1998\)](#) argue that a natural candidate is the debt level, since banks will be reluctant to lend money to debt-heavy firms.

Application 3: Investment and Financing Constraints

- Growth opportunities has been another important variable to distinguish constrained and unconstrained firms.
- [González et al. \(2005\)](#) argue that firms with ample growth opportunities, internal and external finance are substitutes and then firms' investment decisions are independent of its financial structure.
- A variable that relates growth opportunities is a measure of the Tobin's Q ([Gilchrist and Himmelberg, 1995](#); [González et al., 2005](#); [Hubbard, 1998](#); [Lang et al., 1996](#)).
- Thus, in this paper we consider both measures of financial constraints which divide firms into four regimes.

Application 3: Investment and Financing Constraints

That is, we estimate a variation of [Hansen \(1999\)](#)'s model with four regimes

$$I_{it} = \begin{cases} \mu_i + \theta' X_{it} + \beta_3 CF_{it} + e_{it} & \text{if } D_{it} > \gamma_1, Q_{it} \leq \gamma_2 \\ \mu_i + \theta' X_{it} + \beta_1 CF_{it} + e_{it} & \text{if } D_{it} \leq \gamma_1, Q_{it} \leq \gamma_2 \\ \mu_i + \theta' X_{it} + \beta_4 CF_{it} + e_{it} & \text{if } D_{it} > \gamma_1, Q_{it} > \gamma_2 \\ \mu_i + \theta' X_{it} + \beta_2 CF_{it} + e_{it} & \text{if } D_{it} \leq \gamma_1, Q_{it} > \gamma_2, \end{cases}$$

Where I_{it} is the investment to assets ratio, Q_{it} is the total market value to assets ratio, D_{it} is the long-term debt to assets ratio; CF_{it} is the cash flow to assets ratio, i indexes firms and t indexes time.

Following [Hansen \(1999\)](#), we use a balance panel data of 565 US firms from 1973 to 1987.

Application 3: Investment and Financing Constraints

Table 4: Tests for 4 regimes vs 3 regimes

H_1 : 4 regimes		$\beta_1 \neq \beta_2 \neq \beta_3 \neq \beta_4$					
H_0 : 3 regimes		$\beta_1 = \beta_2$	$\beta_1 = \beta_3$	$\beta_1 = \beta_4$	$\beta_2 = \beta_3$	$\beta_2 = \beta_4$	$\beta_3 = \beta_4$
Static model	$F(\hat{\gamma})$	31.3	18.5	0.6	42.2	19.5	24.0
	p-value	0.000	0.007	0.503	0.000	0.013	0.007

Application 3: Investment and Financing Constraints

Table 5: Tests for 3 regimes vs 2 regimes

H_1 : 3 regimes		$\beta_1 = \beta_4 \neq \beta_2 \neq \beta_3$		
H_0 : 2 regimes		$\beta_1 = \beta_4 = \beta_2$	$\beta_1 = \beta_4 = \beta_3$	$\beta_1 = \beta_4$ and $\beta_2 = \beta_3$
Static model	$F(\hat{\gamma})$	36.1	36.4	43.3
	p -value	0.000	0.003	0.000

Application 3: Investment and Financing Constraints

The point estimate of the thresholds and their asymptotic 90, 95 and 99 percent confidence intervals are reported in the next table.

Table 6: Asymptotic confidence interval in threshold model

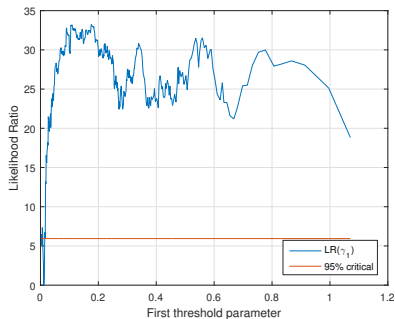
	Threshold estimate (%)	90% confidence interval	95% confidence interval	99% confidence interval
$\hat{\gamma}_1$	0.0125	[0.0043, 0.0152]	[0.0043, 0.0162]	[0.0043, 0.0162]
$\hat{\gamma}_2$	3.5107	[2.9676, 4.3237]	[2.9676, 4.3237]	[2.7540, 4.3237]

Note: Asymptotic critical values are reported in Hansen (2000).

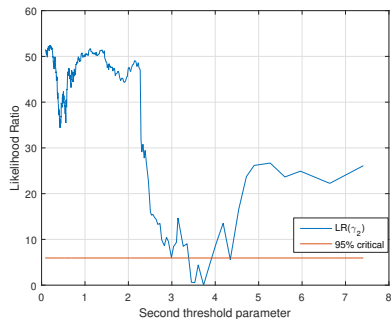
Application 3: Investment and Financing Constraints

Figure 2: Confidence interval construction for thresholds

(a) First threshold parameter



(b) Second threshold parameter



Application 3: Investment and Financing Constraints

The coefficients of primary interest are those expressing the interaction between investment and cash flow.

Table 7: Estimation results - Dependent variable: Investment (I_{it})

Regressor	Estimate	OLS SE
$CF_{it}1(D_{it} \leq \hat{\gamma}_1, Q_{it} > \hat{\gamma}_2)$	-0.010	0.012
$CF_{it}[1(D_{it} > \hat{\gamma}_1, Q_{it} > \hat{\gamma}_2) \text{ or } 1(D_{it} \leq \hat{\gamma}_1, Q_{it} \leq \hat{\gamma}_2)]$	0.035	0.010
$CF_{it}1(D_{it} > \hat{\gamma}_1, Q_{it} \leq \hat{\gamma}_2)$	0.069	0.009
Q_{it}	0.013	0.002
$Q_{it}^2/10^3$	-0.372	0.060
$Q_{it}^3/10^6$	2.684	0.581
D_{it}	0.041	0.006
$Q_{it}D_{it}$	-0.003	0.001

Dynamic Panel Threshold Model

Ramírez-Rondán (2020) extended the static panel model considering dynamics

$$y_{it} = \begin{cases} \mu_i + \beta_1 y_{it-1} + e_{it} & \text{if } q_{it} \leq \gamma \\ \mu_i + \beta_2 y_{it-1} + e_{it} & \text{if } q_{it} > \gamma \end{cases}$$

or in a compact form,

$$y_{it} = \mu_i + \beta_1 y_{it-1} 1(q_{it} \leq \gamma) + \beta_2 y_{it-1} 1(q_{it} > \gamma) + e_{it}, \quad (8)$$

where

- q_{it} is the threshold variable assumed to be exogenous and must have a continuous distribution;
- μ_i is the individual fixed specific effect;
- e_{it} is the error term assumed to be i.i.d normal distributed with mean 0 and variance σ_e^2 ;
- γ is the threshold parameter;
- β_1 and β_2 are the slope parameters.

Dynamic Panel Threshold Model

Two problems in estimating the linear version of model (8) by Maximum Likelihood

The estimation of the slope parameter of the linear version of model (8) is inconsistent as n (number of individuals) grows to infinity.

- Incidental parameters problem in which the number of parameters increases with the number of observations ([Lancaster, 2000](#)).
- Initial condition problem in which the assumption on it plays a crucial role in devising consistent estimates ([Anderson and Hsiao, 1981](#)).

Remark 1: Ignoring initial conditions leads an inconsistency of the MLE of the slope parameter ([Hsiao, Pesaran and Tahmiscioglu, 2002](#)).

Dynamic Panel Threshold Model

Two problems in estimating the linear version of model (8) by Maximum Likelihood

To address the incidental parameter problem we take the first difference to eliminate the individual specific effect in model (8).

To address the initial condition problem, we follow [Hsiao et al. \(2002\)](#) and assume the process has started from a finite period in the past such that $\Delta y_{i1} = \delta + v_{i1}$.

[Ramírez-Rondán \(2020\)](#) propose a maximum likelihood estimation to estimate the threshold and slope parameters is proposed using first difference transformations (for $t = 2, 3, \dots, T$):

$$\Delta y_{it} = \beta_1 \Delta y_{it-1}^*(\gamma) + \beta_2 \Delta y_{it-1}^+(\gamma) + \Delta u_{it}. \quad (9)$$

Dynamic Panel Threshold Model

Estimation

Let $\beta_\delta = (\delta, \beta')'$ and define the matrix $\Delta y_{i,-1}(\gamma)$ as follows

$$\Delta y_{i,-1}(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \Delta y_{i1}^*(\gamma) & \Delta y_{i1}^+(\gamma) \\ 0 & \Delta y_{i2}^*(\gamma) & \Delta y_{i2}^+(\gamma) \\ \vdots & \vdots & \vdots \\ 0 & \Delta y_{iT-1}^*(\gamma) & \Delta y_{iT-1}^+(\gamma) \end{bmatrix};$$

the log-likelihood function can be written as

$$\begin{aligned} \ln L(\beta_\delta, \gamma, \sigma_u^2, \omega) = & -\frac{nT}{2} \ln(2\pi) - \frac{nT}{2} \ln(\sigma_u^2) - \frac{n}{2} \ln[1 + T(\omega - 1)] \\ & - \frac{1}{2} \sum_{i=1}^n \left[(\Delta y_i - \Delta y_{i,-1}(\gamma)\beta_\delta)' \Omega^{-1} (\Delta y_i - \Delta y_{i,-1}(\gamma)\beta_\delta) \right]. \quad (10) \end{aligned}$$

Dynamic Panel Threshold Model

Estimation

We start the estimation procedure by considering a fixed γ . Then, for a given γ , by taking the first-order derivative with respect to β_δ , σ_u^2 and ω , and setting the partial derivative equal to zero gives

$$\widehat{\beta}_\delta(\gamma) = \left(\sum_{i=1}^n \Delta y_{i,-1}(\gamma)' \widehat{\Omega}^*(\gamma)^{-1} \Delta y_{i,-1}(\gamma) \right)^{-1} \left(\sum_{i=1}^n \Delta y_{i,-1}(\gamma)' \widehat{\Omega}^*(\gamma)^{-1} \Delta y_i \right),$$

$$\widehat{\sigma}_u^2(\gamma) = \frac{1}{nT} \sum_{i=1}^n \left[(\Delta y_i - \Delta y_{i,-1}(\gamma) \widehat{\beta}_\delta(\gamma))' \widehat{\Omega}^*(\gamma)^{-1} (\Delta y_i - \Delta y_{i,-1}(\gamma) \widehat{\beta}_\delta(\gamma)) \right],$$

and

$$\widehat{\omega}(\gamma) = \frac{T-1}{T} + \frac{1}{\widehat{\sigma}_u^2(\gamma) n T^2} \sum_{i=1}^n \left[(\Delta y_i - \Delta y_{i,-1}(\gamma) \widehat{\beta}_\delta(\gamma))' \Phi (\Delta y_i - \Delta y_{i,-1}(\gamma) \widehat{\beta}_\delta(\gamma)) \right].$$

Dynamic Panel Threshold Model

Estimation

By plugging the MLE for a given γ , $\hat{\beta}_\delta(\gamma) = (\hat{\delta}(\gamma), \hat{\beta}_1(\gamma), \hat{\beta}_2(\gamma))'$, $\hat{\sigma}_u^2(\gamma)$ and $\hat{\omega}(\gamma)$, into (10); the threshold parameter γ is estimated by maximizing the concentrated log-likelihood function

$$\ln L(\gamma) = -\frac{nT}{2} \ln(2\pi) - \frac{n}{2} \ln |\hat{\Omega}(\gamma)| - \frac{1}{2} \sum_{i=1}^n \Delta \hat{u}_i(\gamma)' \hat{\Omega}(\gamma)^{-1} \Delta \hat{u}_i(\gamma). \quad (11)$$

Then, $\hat{\gamma}$ is obtained by maximizing (11).

Finally, the MLE of slope parameters are $\hat{\beta}_1 = \hat{\beta}_1(\hat{\gamma})$ and $\hat{\beta}_2 = \hat{\beta}_2(\hat{\gamma})$.

Also, the MLE of the remaining parameters that involve the estimation method are $\hat{\delta} = \hat{\delta}(\hat{\gamma})$, $\hat{\sigma}_u^2 = \hat{\sigma}_u^2(\hat{\gamma})$ and $\hat{\omega} = \hat{\omega}(\hat{\gamma})$.

Application 4: Investment and Financing Constraints

- Most economic models also exhibit dynamics
- Lagged investment captures the accelerator effect of investment, whereby past investments have a positive effect on future investments ([Aivazian et al., 2005](#)).

Application 4: Investment and Financing Constraints

We estimate a variation of [Hansen \(1999\)](#)'s threshold model

$$\begin{aligned} I_{it} = & \mu_i + \theta_1 I_{it-1} + \theta_1 Q_{it-1} + \theta_2 Q_{it-1}^2 + \theta_3 Q_{it-1}^3 \\ & + \theta_5 Q_{it-1} D_{it-1} + \theta_4 D_{it-1} + \beta_1' CF_{it-1} 1(q_{it} \leq \gamma) \\ & + \beta_2' CF_{it-1} 1(q_{it} > \gamma) + e_{it}, \end{aligned} \quad (12)$$

where

- I_{it} is the investment to assets ratio,
- I_{it-1} is the lag of investment to assets ratio,
- Q_{it} is the total market value to assets ratio,
- D_{it} is the long-term debt to assets ratio,
- CF_{it} is the cash flow to assets ratio,
- i indexes firms and t indexes time.

Application 4: Investment and Financing Constraints

Table 8: Tests for threshold effects

Test for single threshold	
F_1	28.3
P-value	0.000
(10%, 5%, 1% critical values)	(10.3, 14.8, 26.2)

Application 4: Investment and Financing Constraints

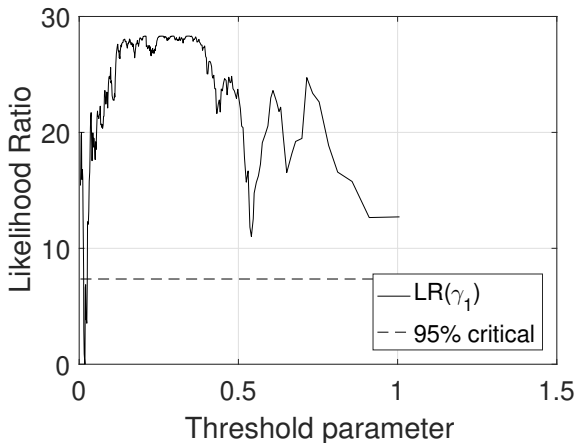
Table 9: Asymptotic confidence interval in threshold model

	Threshold estimate (%)	95% confidence interval
$\hat{\gamma}$	0.0181	[0.0145, 0.0253]

Note: Asymptotic critical values are reported in Hansen (2000).

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Figure 3: Confidence interval construction for threshold



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Table 10: Estimation results - Dependent variable: Investment

Regressor	Coefficient estimate	OLS SE	White SE
I_{it-1}	0.237	0.011	0.070
Q_{it}	0.013	0.001	0.002
$Q_{it}^2/10^3$	-0.366	0.047	0.081
$Q_{it}^3/10^6$	1.047	0.199	0.448
D_{it}	-0.033	0.004	0.009
$Q_{it}D_{it}$	-0.001	0.002	0.005
$CF_{it}1(D_{it} \leq 0.0181)$	0.030	0.006	0.014
$CF_{it}1(0.0181 > D_{it})$	0.059	0.005	0.014

Other Extensions

Let the standard threshold model

$$y_i = \beta_1 x_i 1(q_i \leq \gamma) + \beta_2 x_i 1(q_i > \gamma) + e_i, \quad (13)$$

Using the fact $1(q_i > \gamma) = 1 - 1(q_i \leq \gamma)$, the model can be written as

$$y_i = (\beta_1 - \beta_2) x_i 1(q_i \leq \gamma) + \beta_2 x_i + e_i, \quad (14)$$

If $1(q_i \leq \gamma)$ is replaced by the first order logistic function, $(1 + \exp(-(q_i - \gamma)))^{-1}$, results in a **logistic threshold (LSTAR) model**.

If $1(q_i \leq \gamma)$ is replaced by the exponential function, $1 - \exp(-(q_i - \gamma)^2)$, results in a **exponential threshold (ESTAR) model**.

If $1(q_i \leq \gamma)$ is replaced by a Kernel function, $K(\frac{\gamma - q_i}{h_n})$, results in a **smoothed threshold model**.

Other Extensions

Let the standard threshold model

$$y_i = \begin{cases} \beta_1 x_i + e_i & \text{if } q_i \leq \gamma \\ \beta_2 x_i + e_i & \text{if } q_i > \gamma, \end{cases} \quad (15)$$

If $q_i = \text{time}$, then we have a **linear model with structural change**

$$y_i = \begin{cases} \beta_1 x_i + e_i & \text{if } \text{time} \leq \gamma \\ \beta_2 x_i + e_i & \text{if } \text{time} > \gamma, \end{cases} \quad (16)$$

Let s_t denote an unobservable state variable assuming the value one or zero. A simple **switching Markov model** is given by

$$y_i = \begin{cases} \beta_1 x_i + e_i & \text{if } s_t = 0 \\ \beta_2 x_i + e_i & \text{if } s_t = 1. \end{cases} \quad (17)$$

References

- Hansen, B.E. (1999). Threshold Effects in Non-Dynamic Panels: Estimation, Testing and Inference. *Journal of Econometrics*, 93(2), 345-368.
- Hansen, B.E. (2022b). *Econometrics*. Princeton University Press.
- Lamadrid-Contreras, A. and Ramírez-Rodán, N. (2023). Panel Data Models with Two Threshold Variables. *Studies in Nonlinear Dynamics & Econometrics*, 27(3), 315-333.
- Ramírez-Rodán, N. (2020). Maximum Likelihood Estimation of Dynamic Panel Threshold Models. *Econometric Reviews*, 39(3), 260-276.